

Duplicate Record Check

Task 13 | Data Analytics Track | Level 1, Day 13

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1. Objective

The objective of this task was to find duplicate records in a retail sales dataset, document them, and understand how duplicate detection works. Duplicate records can inflate totals such as revenue and order counts, which leads to wrong business decisions.

2. Tools Used

- Python (Pandas) in Google Colab for detection and cleaning
- Excel format (.xlsx) for the duplicate report

3. Dataset Overview

The dataset is a Retail Sales file containing 1,060 records and 10 columns.

Column	Description
Order_ID	Unique ID of each order (key column)
Order_Date	Date of the order
Customer_ID	ID of the customer
Category / Product	Product category and product name
Quantity / Unit_Price	Units sold and price per unit
Total_Amount	Quantity x Unit_Price
City / Payment_Mode	Store city and payment method (Cash, UPI, Card)

4. Methodology

Duplicates were checked in two ways, as both catch different problems:

- Full-row check: `df.duplicated()` flags rows where every column matches an earlier row.
- Key-column check: `df.duplicated(subset=['Order_ID'], keep=False)` finds repeated Order_IDs and shows all copies, so they can be compared.

Cleaning was done with `drop_duplicates()`: first on full rows, then on Order_ID with `keep='first'`. Every step was recorded in an audit note.

5. Findings

Check	Result
Total records in original file	1,060
Full-row (exact) duplicates	40
Total repeated Order_ID records	60
Conflicting duplicates (same Order_ID, different values)	20
Records after cleaning	1,000

Note: the 60 repeated Order_ID records include the 40 exact duplicates plus 20 conflicting ones.

5.1 Example: Exact Duplicate

Order ORD10007 appears twice with identical values in every column. Such rows are copies and are safe to remove.

Order_ID	Date	Customer	Product	Qty	Price	Total	City	Pay
ORD10007	2025-12-05	CUST216	Jeans	1	99	99	Kolkata	UPI
ORD10007	2025-12-05	CUST216	Jeans	1	99	99	Kolkata	UPI

5.2 Example: Conflicting Duplicate

Order ORD10038 appears twice, but City and Payment_Mode differ. Since one order cannot be placed in two cities with two payment modes, this is a data conflict. The first record was kept, and the other was logged in the report for verification against the source system.

Order_ID	Date	Customer	Product	Total	City	Pay
ORD10038	2025-09-15	CUST371	Headphones	2495	Delhi	Cash
ORD10038	2025-09-15	CUST371	Headphones	2495	Jaipur	Card

6. Impact of Duplicates

Total_Amount summed to 2,872,371 with duplicates and 2,729,249 after cleaning. The 143,122 difference (about 5%) would have been wrongly counted as sales revenue if the duplicates had been left in.

7. Conclusion

Two types of duplicates were found: 40 exact copies and 20 conflicting records sharing an Order_ID. After removing them, 1,000 unique records remained. Checking duplicates on both full rows and key columns is important, because each method catches problems the other misses. Documenting every step in an audit note keeps the cleaning transparent and repeatable.

Learning outcome: I learned how duplicated() and drop_duplicates() work, how the keep and subset parameters change results, and why data should be cleaned before any analysis.

8. Deliverables

- Duplicate report (duplicate_report.xlsx)
- Cleaned copy of the dataset (cleaned_retail_sales.csv)
- Audit note (audit_note.txt)